Firefly Farm: design review graphics

Group 1 · Nimbin, Northern Rivers NSW · one graphic per page, for slides or printing

about 36 ha

property (main block + house parcel)

90–430 m

height range; house about 117 m

36%

of the land steeper than 50%

1,757 mm

mean rain a year (SILO 1995–2025)

11%

of the rain falls Jul–Sep

8.8 h

winter-solstice sun on the house bench

7.8 °C

mean July minimum

40,000 L

existing tank storage

Contents: 2 whole property · 3 house area · 4 rain · 5 temperature · 6 sun and sectors · 7 slope
8 creek-to-ridge profile · 9 water · 10 harvest calendar · 11 phasing · 12 weekly hours

Sources: map data (NSW 5 m elevation, 2019 aerial), SILO, group draft

The whole property: steep bush above a small house bench

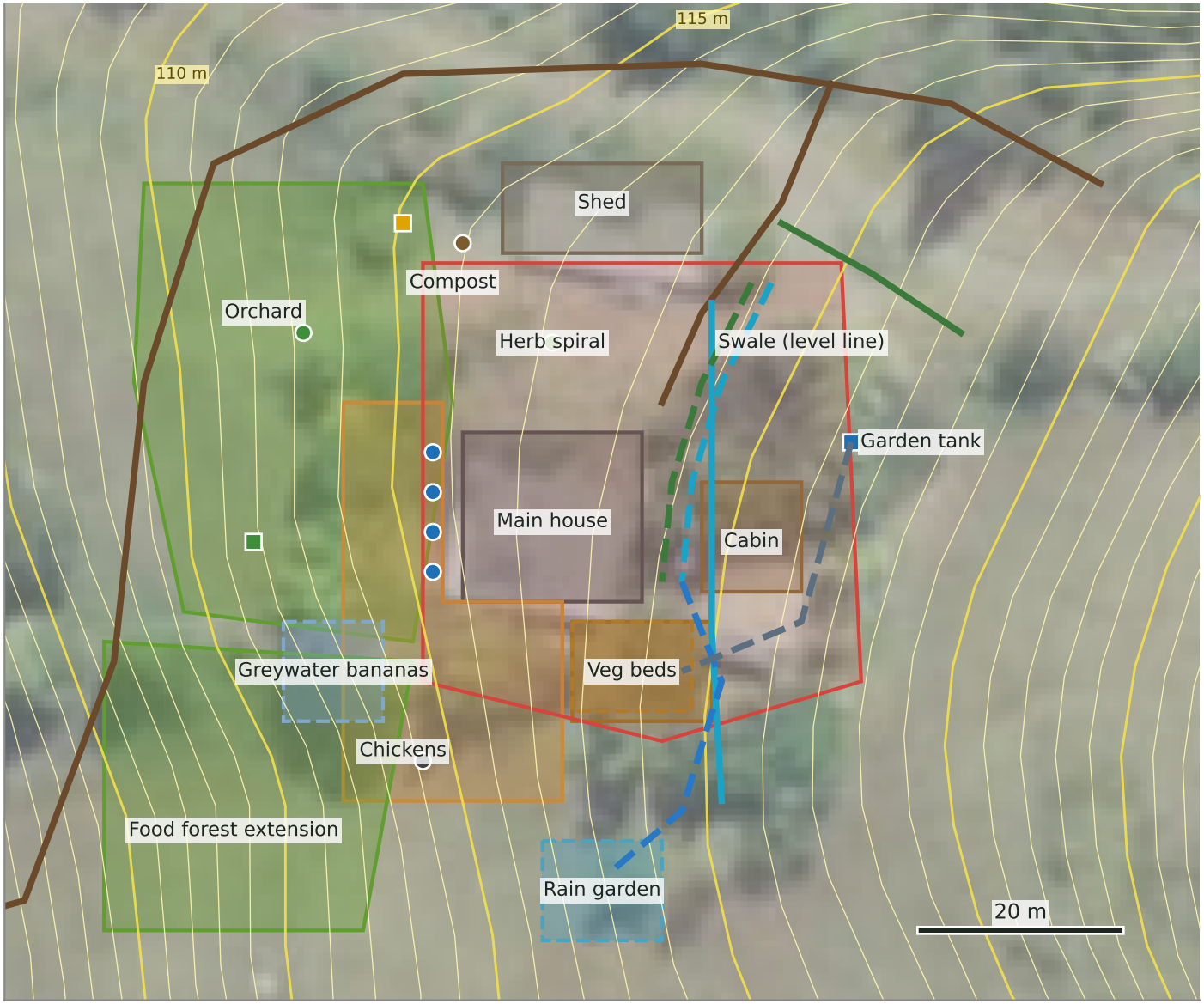
Suggested zones over the 2019 aerial. Red = property outline; blue = creeks; dashed box = the group's house-area boundary



Map data: NSW 5 m elevation, 2019 aerial; zones are suggestions

House area: the group plan plus suggestions

Existing buildings and orchard, the group's plan (garden, chickens, swale, screen) and suggested additions, over 1 m contours



Solid lines: existing and the group plan

Dashed lines, square dots: suggestions

Suggested

- Swale on the level line (about 119 m), spillway south to a rain garden
- Privacy screen under 2 m on the berm
- Garden tank on the high side (about 121 m), gravity to the beds (about 30 kPa)
- Raised or wicking beds after a soil test (arsenic and organochlorines: old banana land)
- Greywater to a banana circle
- Mango guild; native shrubs and macadamia
- Low-fuel zone round the house and cabin

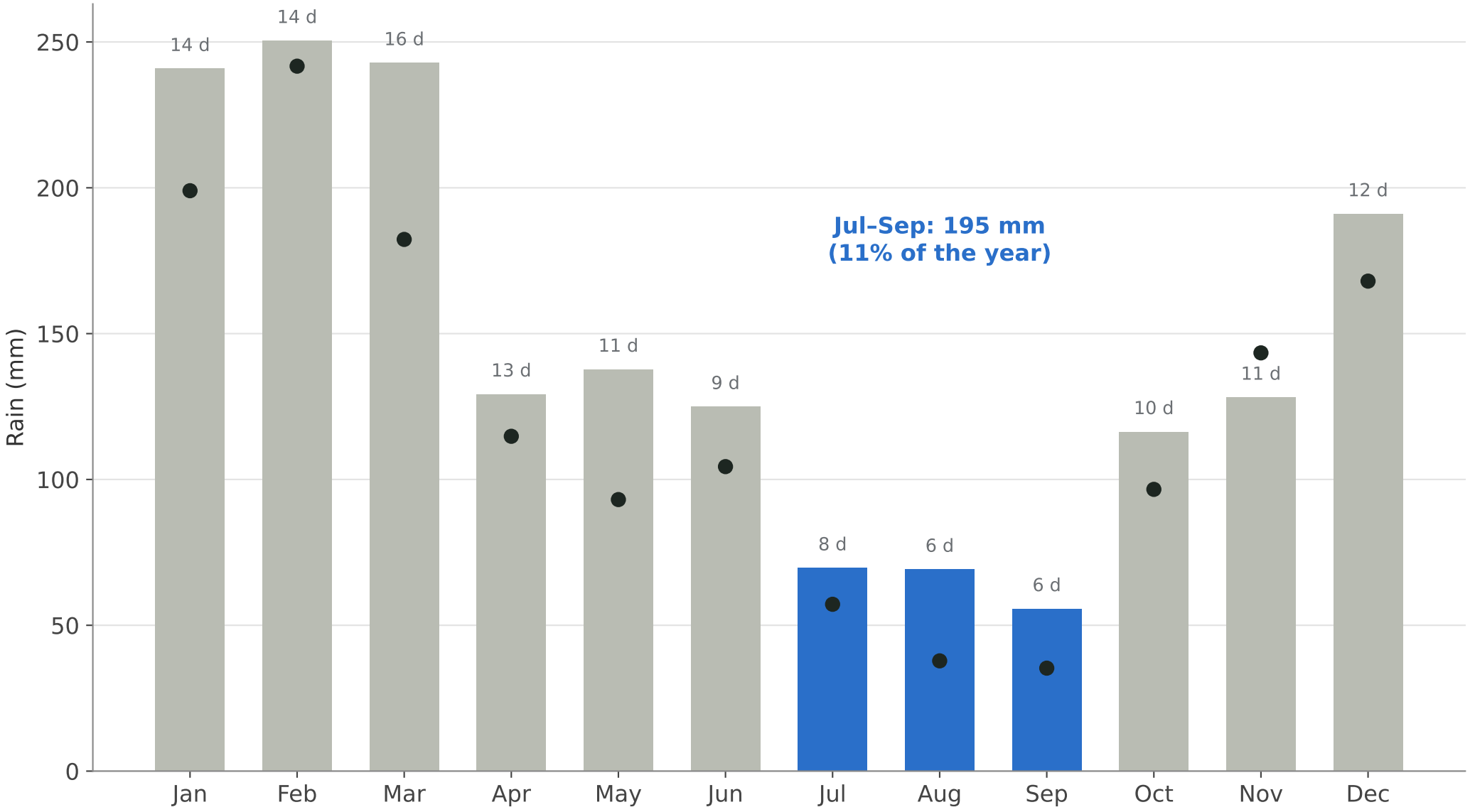
Contours: 1 m (yellow), 5 m labelled

Positions indicative: check on site

Map data: NSW 5 m elevation, 2019 aerial; group draft 23 Oct 2026

Only about a tenth of the year's rain falls in July to September

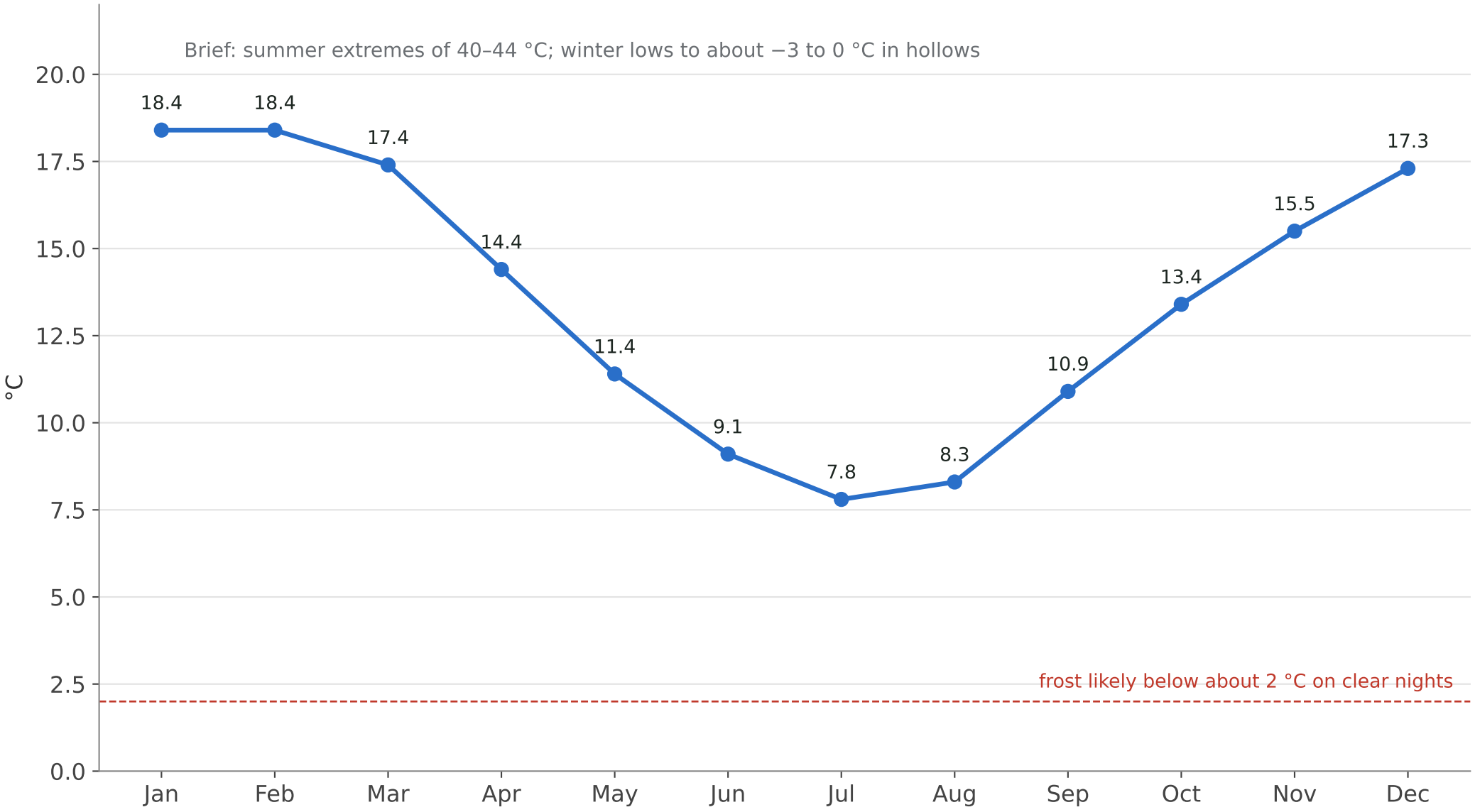
Mean 1,757 mm a year (median year 1,474 mm). Bars = mean, dots = median, numbers = rain days



SILO gridded data at the property (Bureau of Meteorology), 1995–2025

Mild nights: the coldest month averages 7.8 °C

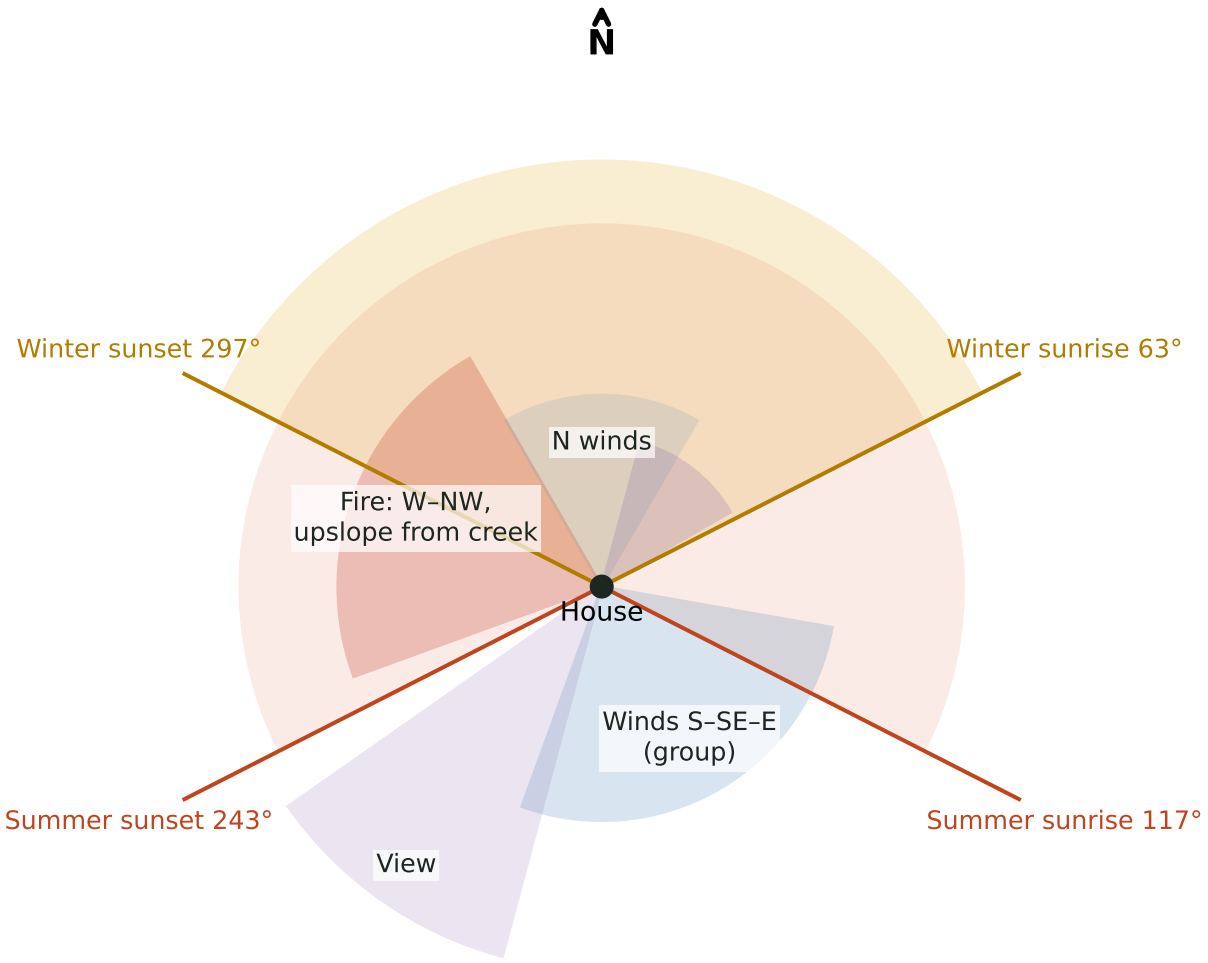
Mean daily minimum temperature. Frost is rare on the house bench; cold air drains to the creek flats

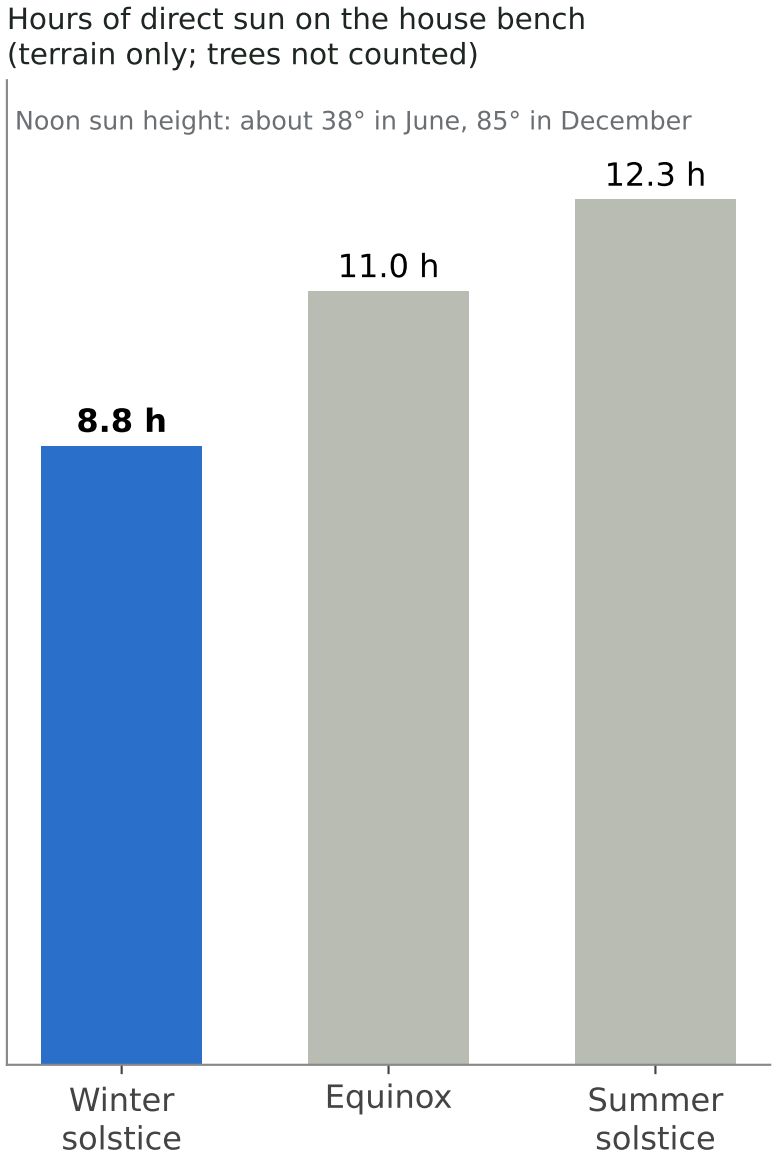


SILO gridded mean minimum temperature, 1995–2025

Sun and sectors at the house

Corrected sun bearings for about 28.6° S (the draft has winter and summer swapped), with wind, fire and view wedges

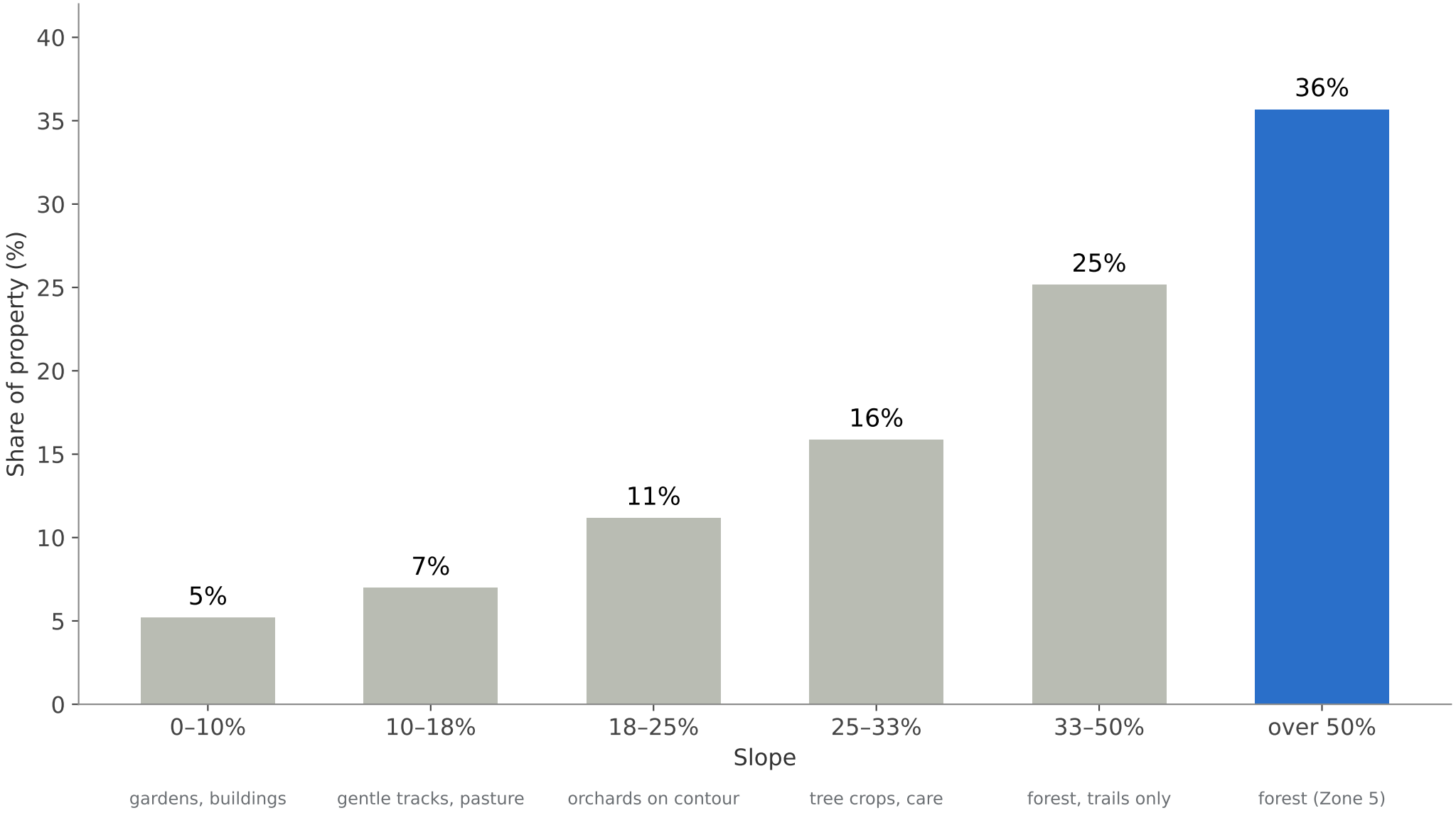




Sun angles calculated for 28.57° S; sun hours from the map (NSW 5 m elevation)

Over a third of the property is steeper than 50%

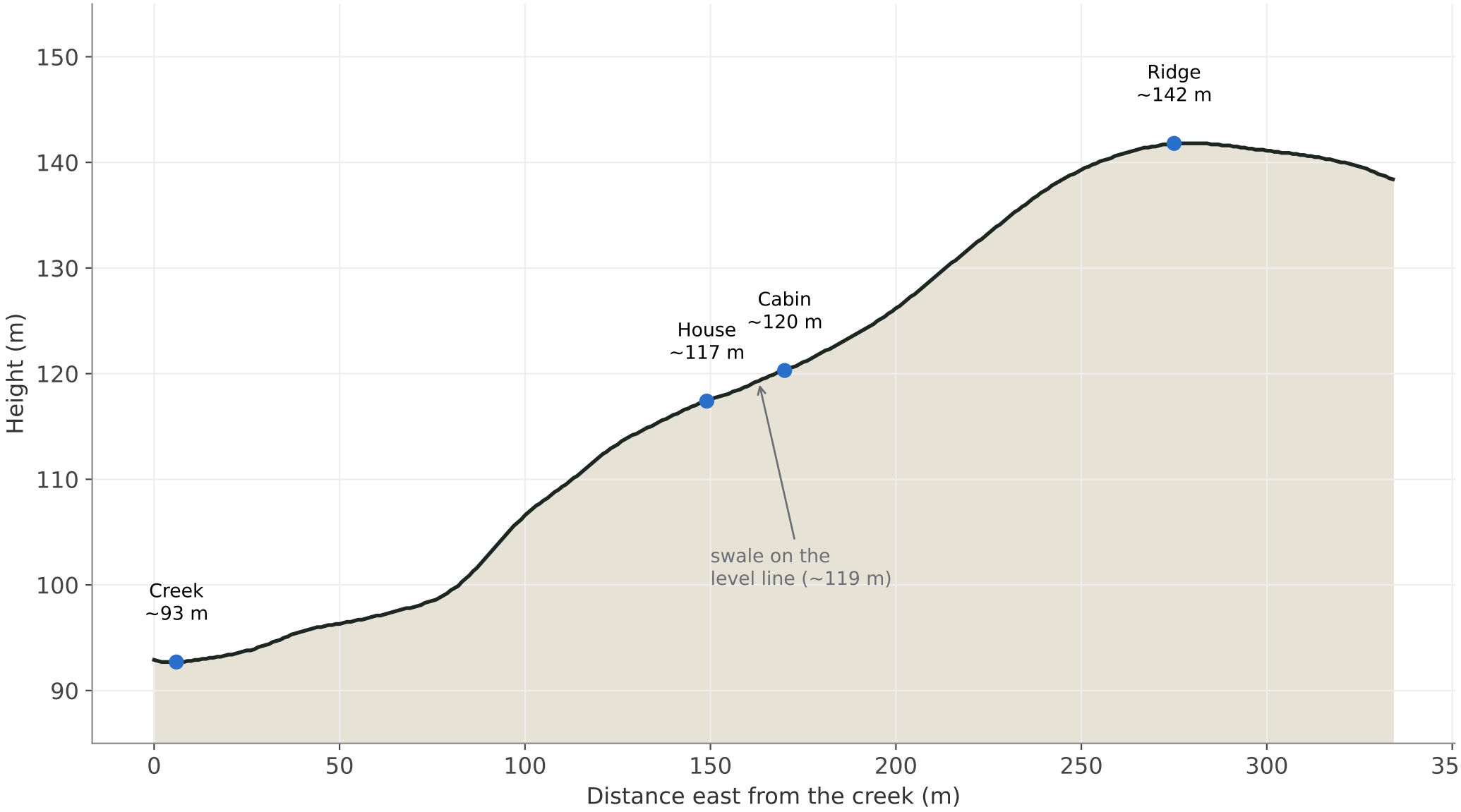
Share of the property in each slope class. Productive use belongs on the gentle ground: the house bench and creek flats



Map data: NSW 5 m elevation (2 m analysis)

Creek to ridge: the house sits on a bench at about 117 m

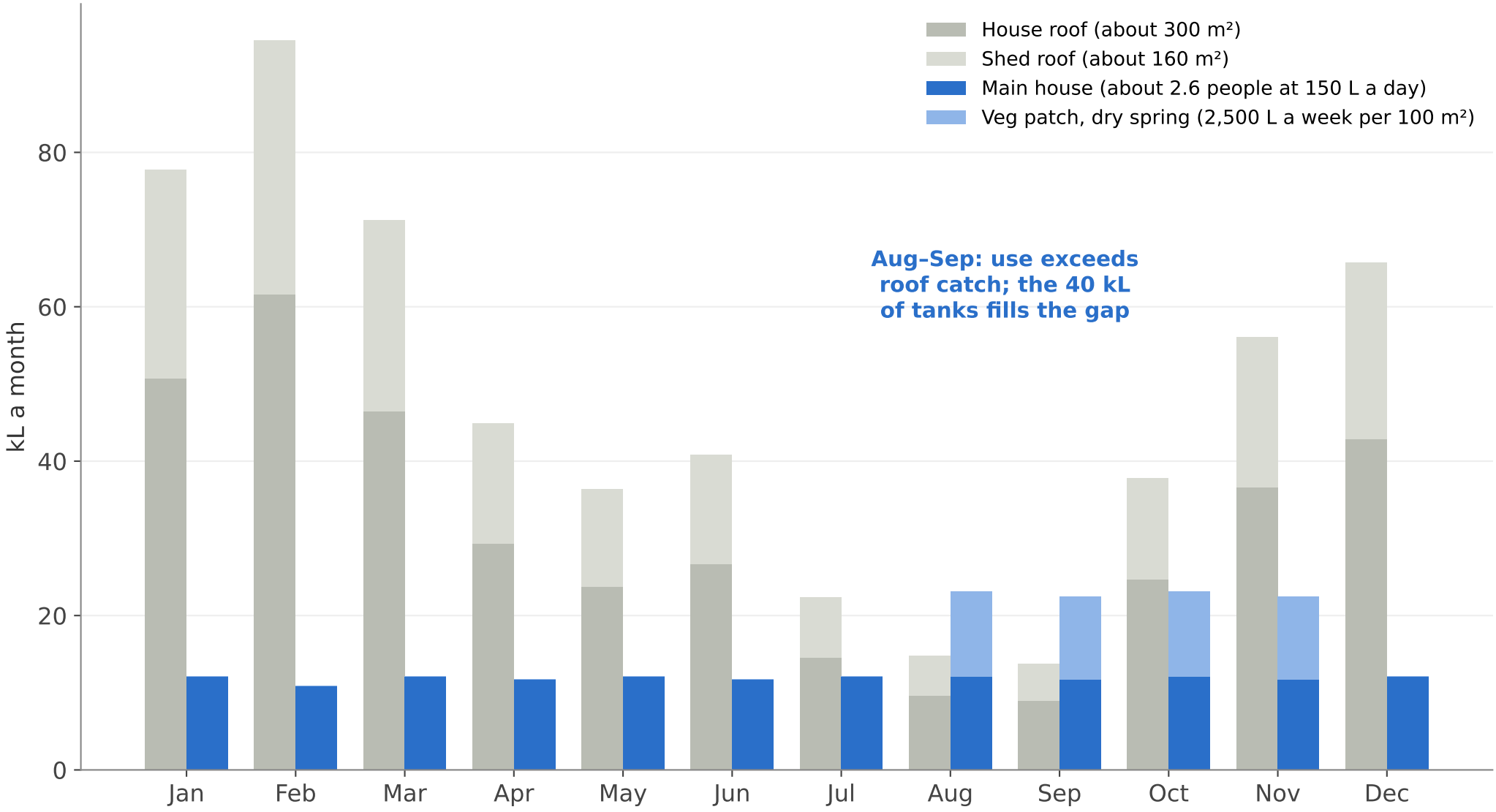
West–east profile through the house (vertical scale exaggerated about 3 times)



Map data: NSW 5 m elevation

Roofs out-catch household use, except in the dry spring

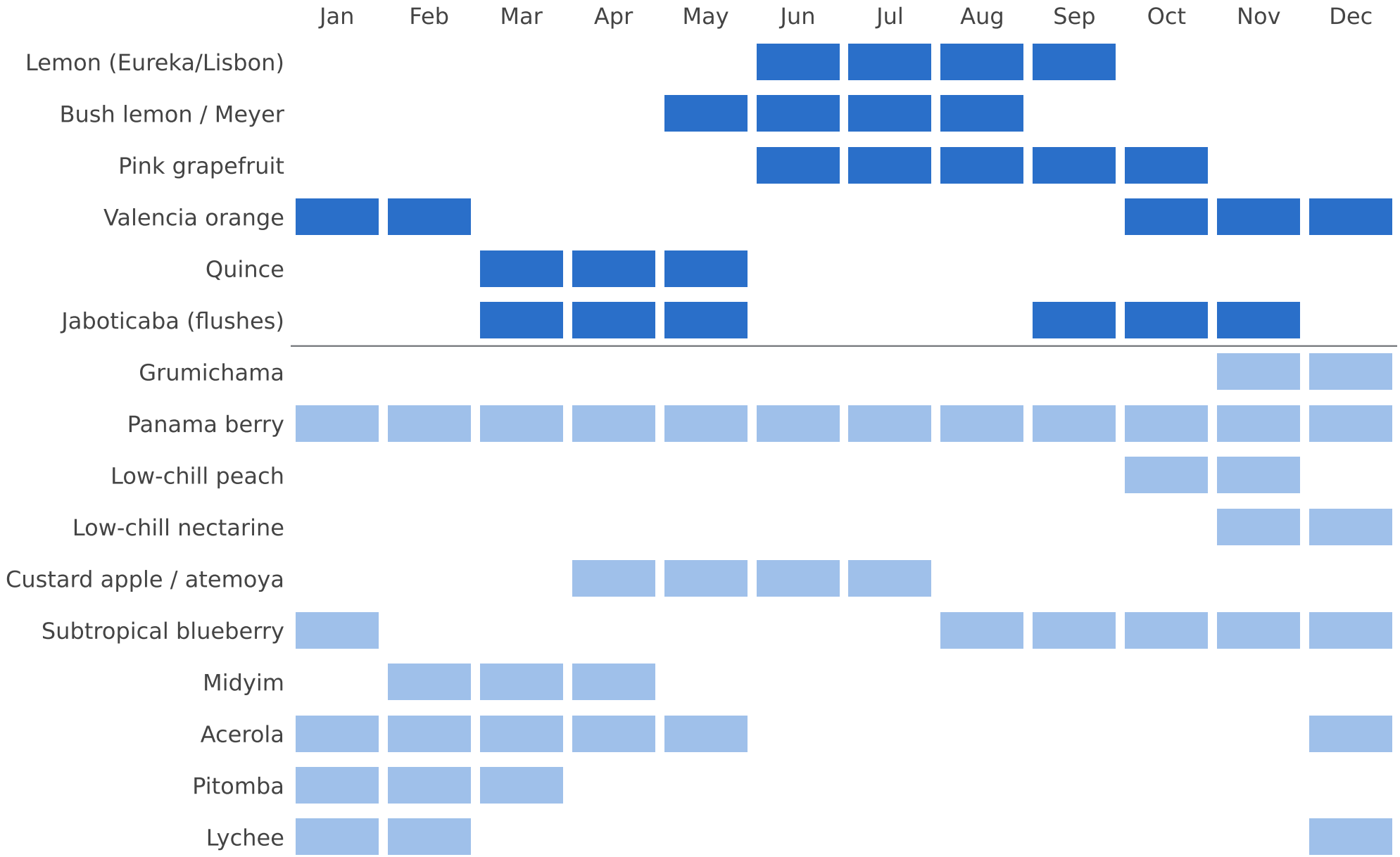
Typical (median) year, kL a month. Supply: house and shed roofs. Use: main house plus a 100 m² vegetable patch



Rain: SILO median 1995–2025; roof sizes approximate; 85% capture

Fruit in every month: the harvest calendar

From the group's species list. Dark = existing trees; light = possible additions



Group draft species list; months as given by the group

Phasing: water, soil and approvals first

A staged plan that fits about 20–30 hours a week

First 3 months

- Watch water in heavy rain
- Soil tests (+ arsenic, organochlorines)
- Council and RFS: cabins, stall
- Gutters, first flush, garden tank
- Herb spiral, compost, 3–4 beds

Year 1

- Chickens and deep-litter run
- Orchard understorey (Mar–May)
- Swale + spillway + screen
- Lantana from the good bush out
- First guest income if approved

Years 2–5

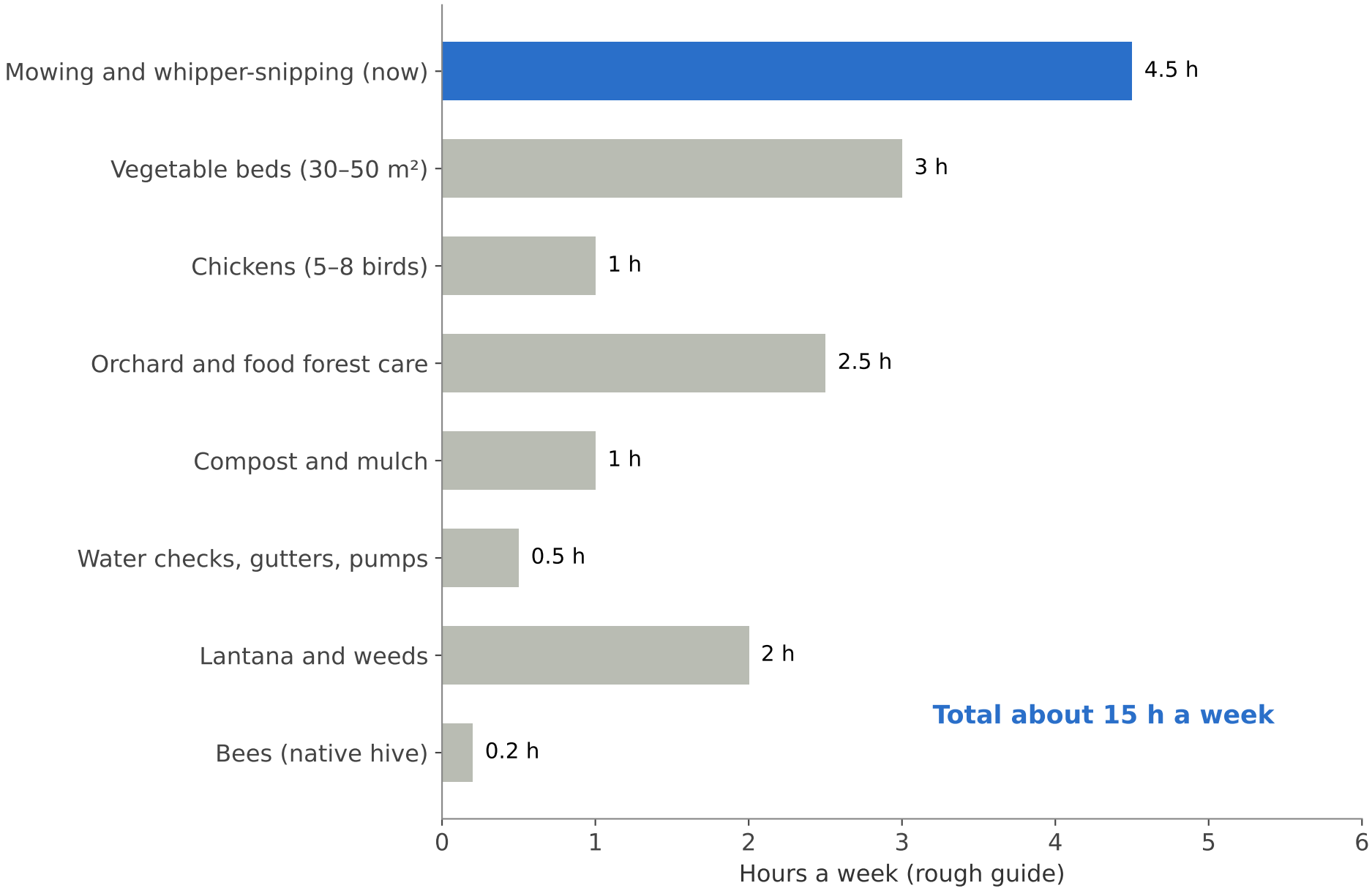
- Food forest + natives, netting
- Cabins if approved
- Bees; a few sheep if time allows
- Pond site from observed flows
- Water-powered creek pump?

Long term

- Syntropic rows on lower slopes
- Bush-food enterprise
- Conservation agreement
- Chosen recreation features

A weekly hours guide: about 15 hours for the core systems

A rough guide to stay inside the brief's 20–30 hours a week. Mulched beds and grazing cut the mowing



Estimates for discussion; the brief gives 4–5 h of mowing now